

HONGYI YANG

hongyiyangzju@gmail.com | +86 156-7411-0092 | github.com/username
Zhejiang University, Hangzhou, China

RESEARCH INTERESTS

Integrated quantum photonics and on-chip quantum light sources; first-principles modelling of optoelectronic devices (NEGF, DFT/DFPT, Wannier functions); hybrid integration of low-dimensional materials on silicon photonic platforms.

EDUCATION

Zhejiang University | B.Eng., Optoelectronic Information Science and Engineering Sep 2023 – Jun 2027 (expected)
Chu Kochen Honors College (Mixed Class, Honors Program)
GPA 4.11 / 4.30 | Rank 4 / 162 (Chu Kochen Honors College), 4 / 102 (School of Optoelectronic Science and Engineering)

PUBLICATIONS (FIRST AUTHOR)

- [1] H. Yang *et al.*, “Sub-Poissonian Electroluminescence from Resonant Inelastic Tunnelling in One-Dimensional Gold Atomic Chains,” *ACS Photonics* — under revision; both referees recommend publication.
- [2] H. Yang *et al.*, “Physically Grounded Neural Networks: From Material Computing to Physics-Constrained Learning,” *Applied Physics Reviews* — under review.

RESEARCH EXPERIENCE

Electrically Driven Non-Classical Light from Gold Atomic Chains Apr 2025 – Present

State Key Laboratory of Extreme Photonics and Instrumentation, Zhejiang University | Advisor: Prof. Haoliang Qian

- Built an end-to-end quantum-transport model of a gold atomic-chain junction: transport Hamiltonian from fully relativistic DFT (Quantum ESPRESSO) projected onto Wannier functions, vibrational modes from DFPT, leads coupled through bulk self-energies.
- Solved the non-equilibrium Green’s function problem beyond the wide-band limit, treating electron–phonon coupling in the self-consistent Born approximation; implemented the full pipeline in Python on HPC clusters.
- Computed current noise as well as current, showing the emitted electroluminescence is sub-Poissonian (Fano factor well below unity) and quantifying how electron–phonon scattering and substitutional doping reshape the photon statistics.
- Coupled the transport model to full-wave simulations of a plasmonic nanocube dimer to link emitter statistics to the local electromagnetic environment.

Strain Engineering of Halide Perovskite Heterostructures Jul 2026 – Oct 2026

Xu Research Group, Stanford University | Visiting Undergraduate Research Intern | Advisor: Prof. Sheng Xu

- Built DFT models of epitaxially strained halide-perovskite heterostructures in Quantum ESPRESSO.
- Quantified how epitaxial compressive strain reshapes the lattice and band structure of the epilayer, giving design guidance for perovskite optoelectronic devices.

HONOURS & AWARDS

National Scholarship — highest national honour for undergraduates in China (top 0.2%)	2023–2024
Huawei Elite Scholarship	2025–2026
OPPO “Benfen” Scholarship — 8 recipients nationwide per year	2024–2025
Top Ten Undergraduates , School of Optoelectronic Science and Engineering, Zhejiang University	Oct 2025
First-Class Scholarship of Zhejiang University ; Excellent Student Title	2023–2024
First Prize , National Undergraduate Optoelectronic Design Competition	2025–2026
First Prize , Zhejiang Provincial College Physics Competition	2024
First Prize , National High School Physics Olympiad (Hunan Province)	2022
3rd Place (Long Jump), 5th Place (Triple Jump) , Zhejiang University Sports Meeting	2023–2025

SKILLS

Programming: Python, PyTorch, C/C++, LaTeX; high-performance computing (SLURM clusters)

Simulation: Custom NEGF transport codes, Quantum ESPRESSO (DFT/DFPT), Wannier90, COMSOL Multiphysics, Zemax

Hardware: SolidWorks, Altium Designer

Languages: Chinese (native), English — TOEFL 103 (R 27 / L 27 / S 25 / W 24)